GOVT 6019: Deriving the Least Squares Line

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0. The problem

We observe n units. For unit i we record two numbers, X_i and Y_i. The units are numbered $i = 1, 2, \ldots, n$.

A line has an intercept and a slope. Write b_0 for a candidate intercept and b_1 for a candidate slope. These are any numbers we might try. For the line with intercept b_0 and slope b_1:

- the **fitted value** for unit i is $\hat{Y}_i = b_0 + b_1 X_i$, the height of the line at X_i;
- the **residual** for unit i is $Y_i - \hat{Y}_i = Y_i - b_0 - b_1 X_i$, the vertical distance from the point to the line. It is positive when the point lies above the line. It is negative when the point lies below.

The **sum of squared residuals** adds up the squared residuals of all n units:

$$\mathrm{SSR}(b_0, b_1) = \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_i)^2 .$$

The data $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$ are fixed numbers. Only b_0 and b_1 vary. So SSR is a function of two inputs, b_0 and b_1. Each candidate line gets one number. A smaller number means the line misses the points by less in total, with each miss measured by its square.

The **ordinary least squares (OLS)** intercept and slope are the pair that makes SSR as small as possible. We write them $\hat{\beta}_0$ and $\hat{\beta}_1$:

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{b_0,\, b_1}{\arg\min}\, \mathrm{SSR}(b_0, b_1).$$

"arg min" means "the input values at which the function is smallest."

This handout finds formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$. Section 1 provides every tool the derivation uses. Sections 2 to 6 derive the formulas. Section 7 rewrites the slope in terms of the correlation. Section 8 checks that every other line has a larger SSR than the OLS line.

Result. If X takes at least two different values,

$$\hat{\beta}_1 = \frac{\widehat{\mathrm{Cov}}(X, Y)}{s_X^2}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

1. Tools

1.1 Sample quantities

$\sum_i$ is shorthand for $\sum_{i=1}^{n}$. Every sum in this handout runs over all n units.

- The **sample means** are $\bar{X} = \frac{1}{n}\sum_i X_i$ and $\bar{Y} = \frac{1}{n}\sum_i Y_i$. Multiplying both sides by n gives a form we use often:
$$\sum_i X_i = n\bar{X}, \qquad \sum_i Y_i = n\bar{Y}.$$
- The **sample variance** of X is $s_X^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X})^2$. The **sample standard deviation** s_X is its square root. s_Y^2 and s_Y are defined the same way, with Y in place of X.
- The **sample covariance** of X and Y is
$$\widehat{\mathrm{Cov}}(X,Y) = \frac{1}{n-1}\sum_i (X_i - \bar{X})(Y_i - \bar{Y}).$$
- The **sample correlation** is $\hat{r} = \dfrac{\widehat{\mathrm{Cov}}(X,Y)}{s_X\, s_Y}$.

1.2 Algebra

- **Distributing.** $c\,(a+b) = ca + cb$ and $c\,(a-b) = ca - cb$.
- **Subtracting a bracket flips every sign inside it.** $a - (b - c) = a - b + c$.
- **Expanding a square.** $(a-b)^2 = a^2 - 2ab + b^2$ and $(a+b)^2 = a^2 + 2ab + b^2$.
- **Doing the same thing to both sides.** If two quantities are equal, they stay equal when we add the same number to both, or multiply or divide both by the same number. Division by zero is the exception: it is never allowed.
- **Zero divided by a nonzero number is zero.** For example, $0/(-2) = 0$.

1.3 Sums

- **Splitting.** The sum of a sum of terms is the sum of the separate sums:
$$\sum_i (a_i + c_i) = \sum_i a_i + \sum_i c_i, \qquad \sum_i (a_i - c_i) = \sum_i a_i - \sum_i c_i.$$
This holds because addition can be done in any order.
- **Constants factor out.** If c is the same in every term, $\sum_i c\, a_i = c \sum_i a_i$. This is distributing, applied to n terms.
- **A sum of n copies of a constant.** $\sum_{i=1}^{n} c = nc$.
- **Deviations from the mean sum to zero.**
$$\sum_i (X_i - \bar{X}) = 0. \tag{D}$$
Proof. Split the sum: $\sum_i (X_i - \bar{X}) = \sum_i X_i - \sum_i \bar{X}$. The first sum is $n\bar{X}$ (Section 1.1). The second is n copies of the constant $\bar{X}$, so it is also $n\bar{X}$. Their difference is $n\bar{X} - n\bar{X} = 0$. ■
For Y, the same three moves give $\sum_i (Y_i - \bar{Y}) = \sum_i Y_i - \sum_i \bar{Y} = n\bar{Y} - n\bar{Y} = 0$.

1.4 Derivatives

What a derivative is. Take a function f of one input b. Its **derivative**, written $\dfrac{d}{db} f(b)$, is the slope of the graph of f at the point b. It is the rate at which $f(b)$ changes as b increases by a very small amount.

- A positive derivative means f is rising at b.
- A negative derivative means f is falling at b.
- A zero derivative means the graph is flat at b.

Four rules. k and c are constants: numbers that do not change when b changes.

1. *A constant has derivative zero.* $\dfrac{d}{db}\,k = 0$. A constant does not change, so its graph is a flat line.
2. *A straight line has derivative equal to its slope.* $\dfrac{d}{db}\,(c\,b) = c$.
3. *A squared input.* $\dfrac{d}{db}\,(c\,b^2) = 2c\,b$.
4. *Sums.* The derivative of a sum is the sum of the derivatives of its terms.

The one derivative this handout needs. For constants a and c,

$$\frac{d}{db}\,(a - c\,b)^2 = -2c\,(a - c\,b). \tag{R}$$

Proof. Expand the square, with $c\,b$ in the role of the second term:

$$(a - c\,b)^2 = a^2 - 2ac\,b + c^2b^2.$$

Take the derivative of each of the three terms.

- a^2 is a constant. By rule 1, its derivative is 0.
- $-2ac\,b$ is a constant, $-2ac$, times b. By rule 2, its derivative is $-2ac$.
- c^2b^2 is a constant, c^2, times b^2. By rule 3, its derivative is $2c^2\,b$.

By rule 4, the derivative of $(a - c\,b)^2$ is the sum of the three: $0 - 2ac + 2c^2b$. Factor out $-2c$. The two pieces are $-2c \cdot a = -2ac$ and $-2c \cdot (-c\,b) = 2c^2b$, so

$$-2ac + 2c^2b = -2c\,(a - c\,b). \qquad \blacksquare$$

The case $c = 1$ comes up first: $\dfrac{d}{db}\,(a - b)^2 = -2\,(a - b)$.

At the lowest point, the slope is zero. Figure 1 shows $f(b) = (b - 3)^2 + 1$, which is smallest at $b = 3$. To the left of 3 the graph falls, so the slope is negative. To the right it rises, so the slope is positive. At $b = 3$ it is flat: the slope is zero. So we can locate the lowest point by finding where the derivative equals zero.

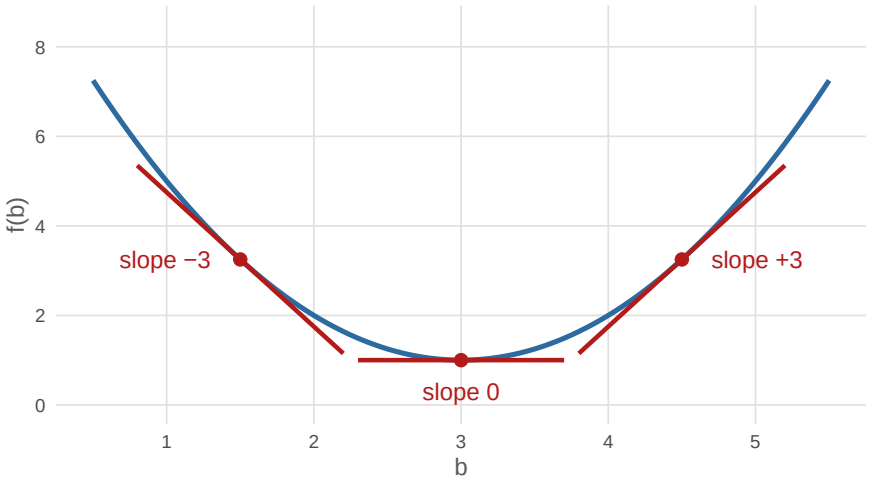


Figure 1: $f(b) = (b-3)^2 + 1$ with its slope drawn at three points. The slope is -3 at $b = 1.5$, 0 at $b = 3$ and $+3$ at $b = 4.5$.

A flat point need not be a lowest point: the top of a hill is flat too. Section 8 checks that the point we find is the lowest one.

1.5 Functions of two inputs

SSR has two inputs, b_0 and b_1. Its **partial derivative** with respect to b_0, written $\dfrac{\partial \text{SSR}}{\partial b_0}$, is the derivative we get by treating b_0 as the only input. Every other letter, including b_1, is held fixed and treated as a constant. The partial derivative with respect to b_1 is defined the same way, with the roles of b_0 and b_1 swapped.

At the lowest point, both partial derivatives are zero. Suppose SSR is smallest at $(\hat{\beta}_0, \hat{\beta}_1)$. Hold b_1 fixed at $\hat{\beta}_1$ and vary only b_0. No value of b_0 gives a smaller SSR than $\hat{\beta}_0$ does, because $(\hat{\beta}_0, \hat{\beta}_1)$ is the overall lowest point. So $b_0 = \hat{\beta}_0$ is the lowest point of this one-input function. Its slope there is zero (Section 1.4). That slope is the partial derivative with respect to b_0. The same argument with the roles swapped shows that the partial derivative with respect to b_1 is zero at $(\hat{\beta}_0, \hat{\beta}_1)$ too.

The two equations "partial derivative $= 0$" are called the **first-order conditions**. Sections 2 to 5 solve them.

2. The first condition: the intercept

Step 1: one term. Look at unit i's term, $(Y_i - b_0 - b_1 X_i)^2$. For the partial derivative with respect to b_0, Y_i, b_1 and X_i are constants. Group them as $a = Y_i - b_1 X_i$. The term is then $(a - b_0)^2$. Rule (R) with $c = 1$ gives its derivative:

$$-2\,(a - b_0) = -2\,(Y_i - b_1 X_i - b_0) = -2\,(Y_i - b_0 - b_1 X_i).$$

The last step only reorders the subtraction.

Step 2: all terms. SSR is a sum of n such terms. By rule 4, its derivative is the sum of the n derivatives from Step 1. The constant -2 factors out of the sum:

$$\frac{\partial \,\mathrm{SSR}}{\partial b_0} = \sum_i -2\,(Y_i - b_0 - b_1 X_i) = -2\sum_i (Y_i - b_0 - b_1 X_i).$$

Step 3: set it to zero at the lowest point. At $(b_0, b_1) = (\hat{\beta}_0, \hat{\beta}_1)$, this partial derivative is zero (Section 1.5):

$$-2\sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0.$$

Step 4: divide both sides by -2. The right side stays 0:

$$\sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0. \tag{1}$$

$Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ is unit i's residual from the OLS line. So equation (1) says the OLS residuals sum to zero.

Step 5: split the sum.

$$\sum_i Y_i - \sum_i \hat{\beta}_0 - \sum_i \hat{\beta}_1 X_i = 0.$$

Step 6: simplify each sum. The first sum is $n\bar{Y}$ (Section 1.1). The second is n copies of the constant $\hat{\beta}_0$, so it is $n\hat{\beta}_0$. In the third, the constant $\hat{\beta}_1$ factors out, leaving $\hat{\beta}_1 \sum_i X_i = \hat{\beta}_1\, n\bar{X}$:

$$n\bar{Y} - n\hat{\beta}_0 - n\hat{\beta}_1 \bar{X} = 0.$$

Step 7: divide both sides by n.

$$\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X} = 0.$$

Step 8: solve for the intercept. Add $\hat{\beta}_0$ to both sides:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}. \tag{2}$$

■

Equation (2) gives the intercept once we know the slope. Rearranged, it says $\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$: the OLS line passes through the point $(\bar{X}, \bar{Y})$.

3. The second condition: the slope

Step 1: one term. For the partial derivative with respect to b_1, Y_i, b_0 and X_i are constants. Group them as $a = Y_i - b_0$ and $c = X_i$. Unit i's term is then $(a - c\,b_1)^2$. Rule (R) gives its derivative:

$$-2c\,(a - c\,b_1) = -2X_i\,(Y_i - b_0 - b_1 X_i).$$

Step 2: all terms. By rule 4, sum the n derivatives. The constant -2 factors out:

$$\frac{\partial \,\mathrm{SSR}}{\partial b_1} = -2\sum_i X_i\,(Y_i - b_0 - b_1 X_i).$$

Step 3: set it to zero at the lowest point.

$$-2\sum_i X_i\,(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0.$$

Step 4: divide both sides by -2.

$$\sum_i X_i\,(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0. \tag{3}$$

Step 5: substitute the intercept. Equation (2) says $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X}$. Put this in place of $\hat{\beta}_0$ inside the bracket, then remove the inner bracket. Subtracting it flips both signs inside (Section 1.2):

$$Y_i - (\bar{Y} - \hat{\beta}_1\bar{X}) - \hat{\beta}_1 X_i = Y_i - \bar{Y} + \hat{\beta}_1\bar{X} - \hat{\beta}_1 X_i.$$

Group the Y terms and the $\hat{\beta}_1$ terms. Factor $-\hat{\beta}_1$ out of the last two, since $\hat{\beta}_1\bar{X} - \hat{\beta}_1 X_i = -\hat{\beta}_1(X_i - \bar{X})$:

$$Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i = (Y_i - \bar{Y}) - \hat{\beta}_1(X_i - \bar{X}).$$

Equation (3) becomes

$$\sum_i X_i\left[(Y_i - \bar{Y}) - \hat{\beta}_1(X_i - \bar{X})\right] = 0.$$

Step 6: distribute and split. Multiply X_i into the bracket. Then split the sum into two sums:

$$\sum_i X_i(Y_i - \bar{Y}) - \sum_i \hat{\beta}_1\,X_i(X_i - \bar{X}) = 0.$$

In the second sum, the constant $\hat{\beta}_1$ factors out:

$$\sum_i X_i(Y_i - \bar{Y}) - \hat{\beta}_1\sum_i X_i(X_i - \bar{X}) = 0.$$

Step 7: move the $\hat{\beta}_1$ term to the right. Add $\hat{\beta}_1\sum_i X_i(X_i - \bar{X})$ to both sides:

$$\sum_i X_i(Y_i - \bar{Y}) = \hat{\beta}_1\sum_i X_i(X_i - \bar{X}). \tag{4}$$

4. Replacing X_i with $X_i - \bar{X}$

Equation (4) can be solved for $\hat{\beta}_1$ as it stands. This section rewrites both sums so the answer comes out in terms of the sample covariance and variance (Section 1.1).

Claim A. $\sum_i X_i(Y_i - \bar{Y}) = \sum_i (X_i - \bar{X})(Y_i - \bar{Y})$.

Step 1: distribute. Multiply $(Y_i - \bar{Y})$ by each part of $(X_i - \bar{X})$:

$$(X_i - \bar{X})(Y_i - \bar{Y}) = X_i(Y_i - \bar{Y}) - \bar{X}\,(Y_i - \bar{Y}).$$

Step 2: sum over units and split.

$$\sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_i X_i(Y_i - \bar{Y}) - \sum_i \bar{X}\,(Y_i - \bar{Y}).$$

Step 3: the second sum is zero. $\bar{X}$ is the same in every term, so it factors out. The remaining sum is zero by (D), applied to Y:

$$\sum_i \bar{X}\,(Y_i - \bar{Y}) = \bar{X} \sum_i (Y_i - \bar{Y}) = \bar{X} \cdot 0 = 0.$$

Step 4: put it together. Subtracting zero changes nothing:

$$\sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_i X_i (Y_i - \bar{Y}).$$

■

Claim B. $\sum_i X_i (X_i - \bar{X}) = \sum_i (X_i - \bar{X})^2$.

The proof is Claim A's, with X in place of Y throughout.

Step 1: distribute.

$$(X_i - \bar{X})(X_i - \bar{X}) = X_i (X_i - \bar{X}) - \bar{X}\,(X_i - \bar{X}).$$

The left side is $(X_i - \bar{X})^2$.

Step 2: sum over units and split.

$$\sum_i (X_i - \bar{X})^2 = \sum_i X_i (X_i - \bar{X}) - \sum_i \bar{X}\,(X_i - \bar{X}).$$

Step 3: the second sum is zero. Factor out $\bar{X}$ and use (D):

$$\sum_i \bar{X}\,(X_i - \bar{X}) = \bar{X} \sum_i (X_i - \bar{X}) = \bar{X} \cdot 0 = 0.$$

Step 4: put it together.

$$\sum_i (X_i - \bar{X})^2 = \sum_i X_i (X_i - \bar{X}).$$

■

Equation (4), rewritten. Replace the left side of (4) using Claim A and the sum on the right using Claim B:

$$\sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = \hat{\beta}_1 \sum_i (X_i - \bar{X})^2. \tag{5}$$

5. Solving for the slope

Step 1: check that we may divide. To isolate $\hat{\beta}_1$ in (5), we divide both sides by $\sum_i (X_i - \bar{X})^2$. Division by zero is never allowed, so this sum must not be zero.

Each term $(X_i - \bar{X})^2$ is a square, so it is zero or positive. A sum of terms that are zero or positive equals zero only when every term is zero. Every term is zero only when $X_i = \bar{X}$ for every unit, so that all the X_i are equal. So **if X takes at least two different values, the sum is positive**. We may then divide. If every unit has the same X, no slope can be estimated. Every line through $(\bar{X}, \bar{Y})$ then fits equally well.

Step 2: divide.

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}. \tag{6}$$

Step 3: divide top and bottom by $n-1$. Dividing the top and the bottom of a fraction by the same nonzero number leaves its value unchanged. Here $n \geq 2$, because X takes at least two values, so $n-1$ is not zero:

$$\hat{\beta}_1 = \frac{\frac{1}{n-1}\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\frac{1}{n-1}\sum_i (X_i - \bar{X})^2}.$$

Step 4: name the pieces. By the definitions in Section 1.1, the top is $\widehat{\text{Cov}}(X, Y)$ and the bottom is s_X^2:

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2}. \tag{7}$$

■

6. The OLS coefficients

Equations (7) and (2) together give the result stated in Section 0.

Bivariate ordinary least squares (OLS). If X takes at least two different values,

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Compute $\hat{\beta}_1$ first. Then put it into the formula for $\hat{\beta}_0$.

7. Slope and correlation

Result. If X and Y each take at least two different values,

$$\hat{\beta}_1 = \hat{r}\,\frac{s_Y}{s_X}.$$

s_Y must not be zero, because we divide by it in Step 2. By the argument of Section 5, Step 1 with Y in place of X, $s_Y > 0$ exactly when Y takes at least two different values.

Step 1: the OLS slope. Start from (7):

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2}.$$

Step 2: multiply by s_Y/s_Y. $s_Y/s_Y = 1$. Multiplying by 1 changes nothing. Write $s_X^2 = s_X \cdot s_X$:

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X \cdot s_X} \cdot \frac{s_Y}{s_Y}.$$

Multiplying two fractions multiplies their tops and their bottoms. So this is one fraction, with two factors on top and three on the bottom:

$$\hat{\beta}_1 = \frac{\widehat{\mathrm{Cov}}(X,Y) \cdot s_Y}{s_X \cdot s_X \cdot s_Y}.$$

Split it back into two fractions, paired differently. $\widehat{\mathrm{Cov}}(X,Y)$ goes over one s_X and the s_Y. The top s_Y goes over the other s_X:

$$\hat{\beta}_1 = \frac{\widehat{\mathrm{Cov}}(X,Y)}{s_X\, s_Y} \cdot \frac{s_Y}{s_X}.$$

Step 3: name the first fraction. By the definition in Section 1.1, the first fraction is $\hat{r}$:

$$\hat{\beta}_1 = \hat{r}\, \frac{s_Y}{s_X}.$$

■

s_Y and s_X are both positive, so $\hat{\beta}_1$ and $\hat{r}$ always have the same sign. Multiplying both sides by s_X gives $s_X \cdot \hat{\beta}_1 = \hat{r} \cdot s_Y$.

What this says about fitted values. Take one unit with $X_i = x$ and another with $X_i = x + s_X$, one standard deviation of X higher. Their fitted values are $\hat{\beta}_0 + \hat{\beta}_1 x$ and $\hat{\beta}_0 + \hat{\beta}_1(x + s_X)$. Subtract the first from the second. $\hat{\beta}_0$ and $\hat{\beta}_1 x$ cancel:

$$\Big(\hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_1 s_X\Big) - \Big(\hat{\beta}_0 + \hat{\beta}_1 x\Big) = \hat{\beta}_1 s_X = \hat{r}\, s_Y.$$

If $\hat{r} > 0$, the unit with the larger X has a fitted value $\hat{r}\, s_Y$ higher. If $\hat{r} < 0$, its fitted value is lower, by $|\hat{r}|\, s_Y$. In both cases the gap is $|\hat{r}|$ standard deviations of Y.

8. The formulas give the lowest SSR

Sections 2 to 5 assumed that SSR has a lowest point. They showed that this point must satisfy formulas (2) and (7). Two things remain open. Section 1.4 warned that a flat point could be the top of a hill instead of the bottom. The argument also never showed that a lowest point exists. This section closes both gaps. From here on, $\hat{\beta}_1$ and $\hat{\beta}_0$ *are defined by* formulas (7) and (2). We show that this pair has a smaller SSR than every other line.

> **Result.** If X takes at least two different values, then for every pair (b_0, b_1) other than $(\hat{\beta}_0, \hat{\beta}_1)$,
>
> $$\mathrm{SSR}(b_0, b_1) > \mathrm{SSR}(\hat{\beta}_0, \hat{\beta}_1).$$

Write $\hat{\varepsilon}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ for unit i's residual from the OLS line. The proof needs two facts about these residuals:

$$\sum_i \hat{\varepsilon}_i = 0, \qquad \sum_i X_i\, \hat{\varepsilon}_i = 0.$$

These are equations (1) and (3). Sections 2 and 3 derived them from the assumption that $(\hat{\beta}_0, \hat{\beta}_1)$ is a lowest point. We cannot use that assumption here. So we check both facts directly from formulas (2) and (7).

Check 1: the residuals sum to zero. Split the sum. Then simplify each piece as in Section 2, Step 6:

$$\sum_i \hat{\varepsilon}_i = \sum_i Y_i - \sum_i \hat{\beta}_0 - \hat{\beta}_1 \sum_i X_i = n\bar{Y} - n\hat{\beta}_0 - n\hat{\beta}_1\bar{X}.$$

Factor out n:

$$\sum_i \hat{\varepsilon}_i = n\left(\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1\bar{X}\right).$$

Formula (2) says $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X}$. Subtracting $\hat{\beta}_0$ from both sides gives $\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1\bar{X} = 0$. So the bracket is zero: $\sum_i \hat{\varepsilon}_i = n \cdot 0 = 0$.

Check 2: the residuals times X_i sum to zero. Section 3, Step 5 used only formula (2) to rewrite each residual:

$$\hat{\varepsilon}_i = (Y_i - \bar{Y}) - \hat{\beta}_1(X_i - \bar{X}).$$

Multiply by X_i, sum and split, as in Section 3, Step 6:

$$\sum_i X_i\,\hat{\varepsilon}_i = \sum_i X_i(Y_i - \bar{Y}) - \hat{\beta}_1 \sum_i X_i(X_i - \bar{X}).$$

Replace each sum using Claims A and B (Section 4):

$$\sum_i X_i\,\hat{\varepsilon}_i = \sum_i (X_i - \bar{X})(Y_i - \bar{Y}) - \hat{\beta}_1 \sum_i (X_i - \bar{X})^2.$$

Formula (7) equals formula (6) (Section 5, Step 3). Multiply both sides of (6) by its denominator, $\sum_i (X_i - \bar{X})^2$:

$$\hat{\beta}_1 \sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \bar{X})(Y_i - \bar{Y}).$$

So the two terms on the right are equal. Their difference is zero: $\sum_i X_i\,\hat{\varepsilon}_i = 0$.

Step 1: write any residual using the OLS residual. Take any candidate line, with intercept b_0 and slope b_1. Let $d_0 = \hat{\beta}_0 - b_0$ and $d_1 = \hat{\beta}_1 - b_1$. These measure how far the candidate's intercept and slope are from OLS. Then

$$\hat{\varepsilon}_i + d_0 + d_1 X_i = (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) + (\hat{\beta}_0 - b_0) + (\hat{\beta}_1 X_i - b_1 X_i).$$

$\hat{\beta}_0$ appears once with each sign, so it cancels. So does $\hat{\beta}_1 X_i$. What remains is

$$\hat{\varepsilon}_i + d_0 + d_1 X_i = Y_i - b_0 - b_1 X_i.$$

The candidate's residual equals the OLS residual plus $d_0 + d_1 X_i$.

Step 2: square. Expand the square of a sum (Section 1.2), with $\hat{\varepsilon}_i$ in the role of a and $d_0 + d_1 X_i$ in the role of b:

$$(Y_i - b_0 - b_1 X_i)^2 = \hat{\varepsilon}_i^2 + 2\,\hat{\varepsilon}_i\,(d_0 + d_1 X_i) + (d_0 + d_1 X_i)^2.$$

Step 3: sum over units and split. The left side sums to $\mathrm{SSR}(b_0, b_1)$. The first term on the right sums to $\sum_i \hat{\varepsilon}_i^2 = \mathrm{SSR}(\hat{\beta}_0, \hat{\beta}_1)$:

$$\mathrm{SSR}(b_0, b_1) = \mathrm{SSR}(\hat{\beta}_0, \hat{\beta}_1) + 2\sum_i \hat{\varepsilon}_i\,(d_0 + d_1 X_i) + \sum_i (d_0 + d_1 X_i)^2.$$

Step 4: the middle sum is zero. Distribute $\hat{\varepsilon}_i$ and split. d_0 and d_1 are the same for every unit, so they factor out:

$$\sum_i \hat{\varepsilon}_i \, (d_0 + d_1 X_i) = d_0 \sum_i \hat{\varepsilon}_i + d_1 \sum_i X_i \, \hat{\varepsilon}_i = d_0 \cdot 0 + d_1 \cdot 0 = 0.$$

The two zeros are Checks 1 and 2. So

$$\mathrm{SSR}(b_0, b_1) = \mathrm{SSR}(\hat{\beta}_0, \hat{\beta}_1) + \sum_i (d_0 + d_1 X_i)^2. \tag{8}$$

Step 5: the extra sum is never negative. Each term $(d_0 + d_1 X_i)^2$ is a square, so it is zero or positive. By (8), no candidate line has a smaller SSR than the OLS line.

Step 6: only OLS reaches the minimum. A candidate has the same SSR as the OLS line only if the extra sum in (8) is zero. That requires every term to be zero: $d_0 + d_1 X_i = 0$ for every unit. X takes at least two different values, so pick two units, j and k, with $X_j \neq X_k$. Then

$$d_0 + d_1 X_j = 0 \quad \text{and} \quad d_0 + d_1 X_k = 0.$$

Subtract the second equation from the first: $d_1(X_j - X_k) = 0$. Since $X_j - X_k$ is not zero, divide by it: $d_1 = 0$. Put $d_1 = 0$ into the first equation: $d_0 = 0$. So $b_0 = \hat{\beta}_0$ and $b_1 = \hat{\beta}_1$. Every other line has a strictly larger SSR. ■

The OLS line is therefore the only line with the smallest sum of squared residuals.